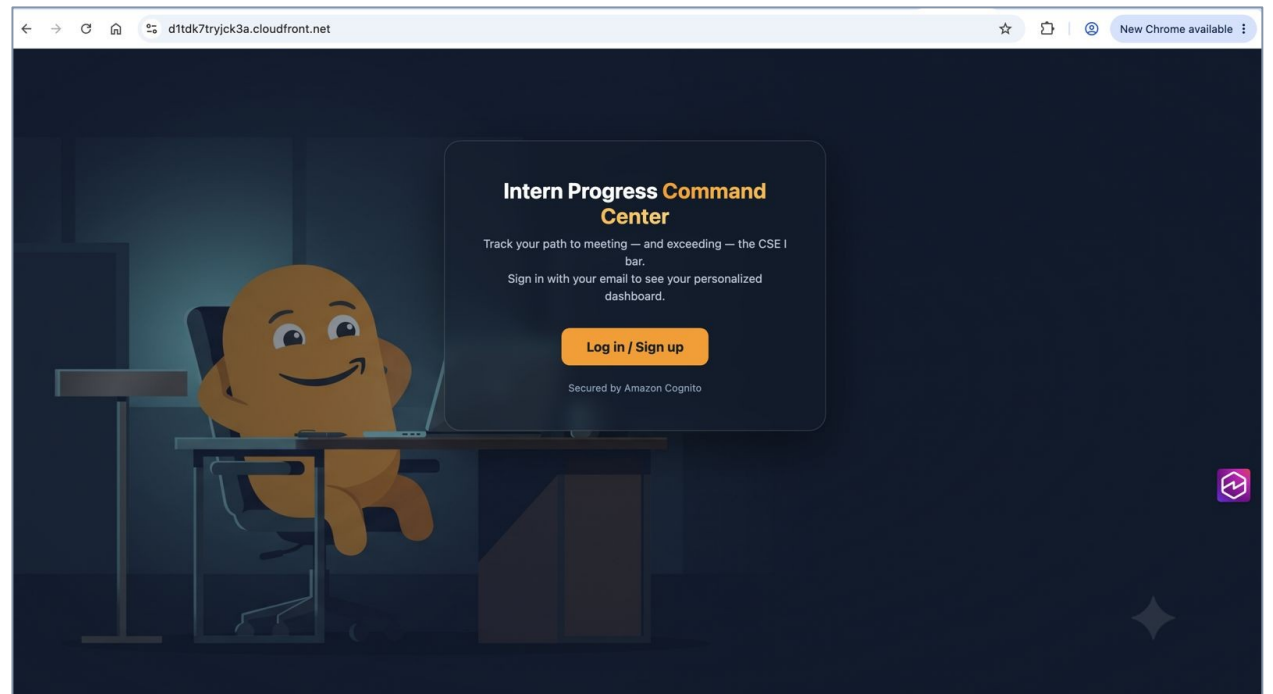


Intern Progress Command Center

Deployed on Amazon ECS (Fargate) · Larryking Akere Mokom (mokoml) · AMER CSE I Capstone

Why I built it

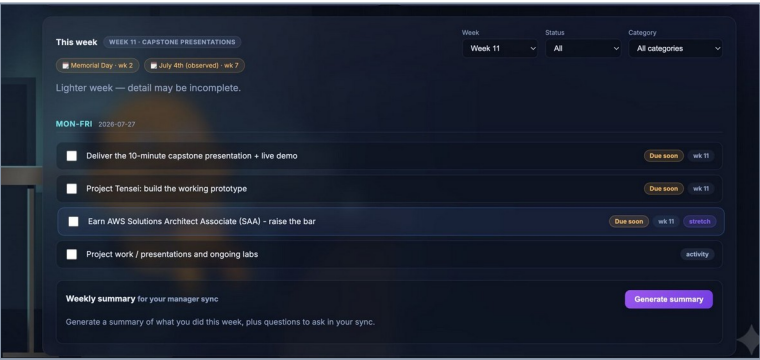
- I'm the customer here — an intern going through this capstone myself.
- So before writing code, I asked other interns what would actually help.
- Their progress was scattered — no single place to see where they stood.
- They wanted reminders so a task never slips by unnoticed.
- And real suggestions on how to raise the bar, not just log work.



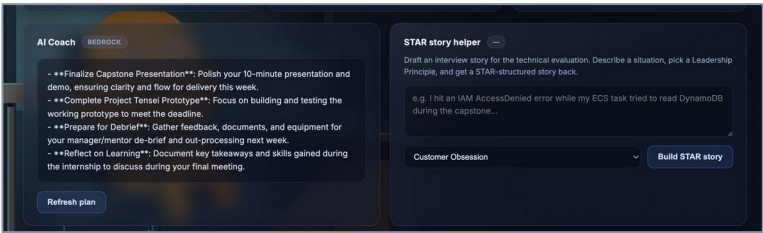
What it does



One dashboard — where you stand

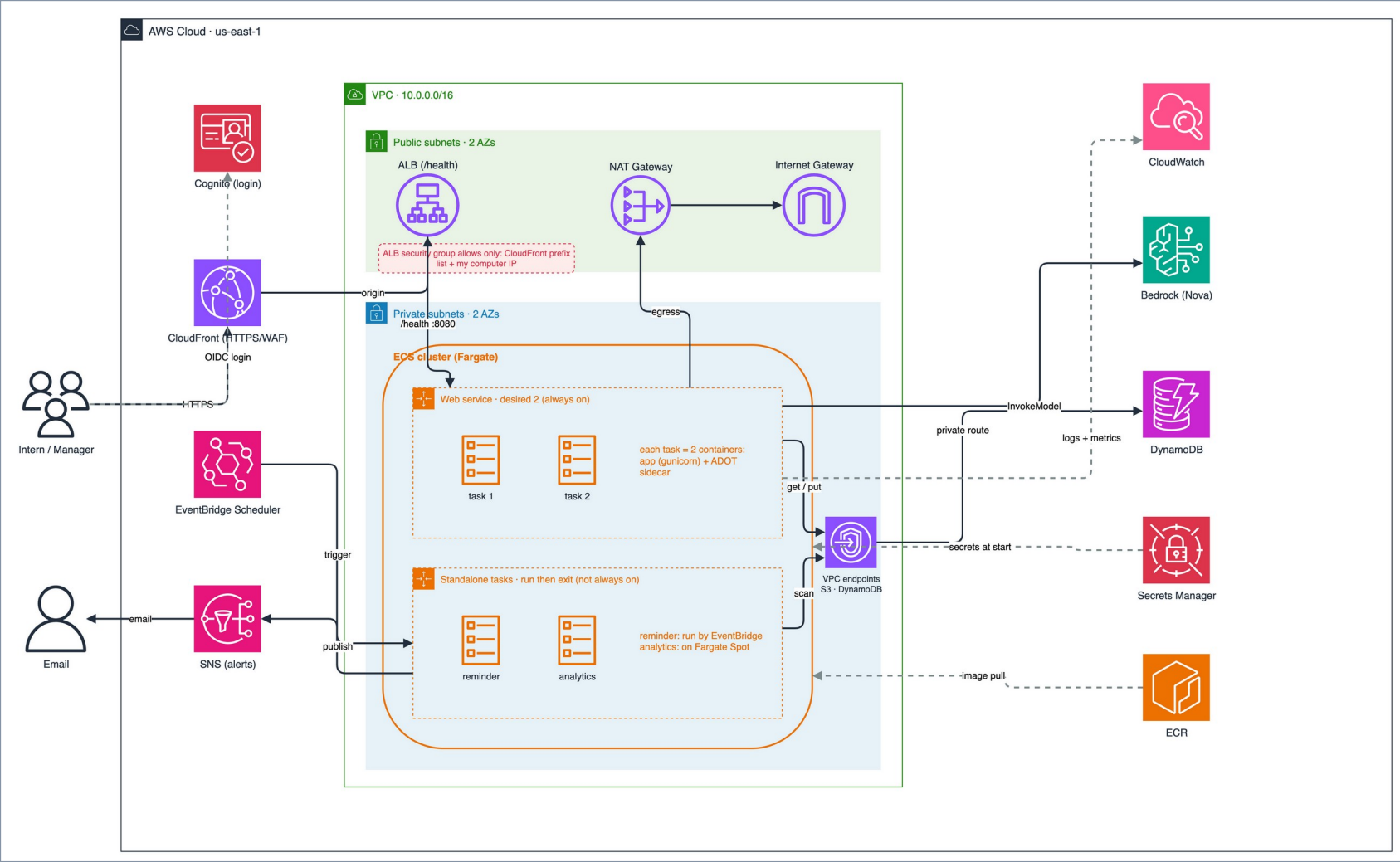


Your week — nothing slips



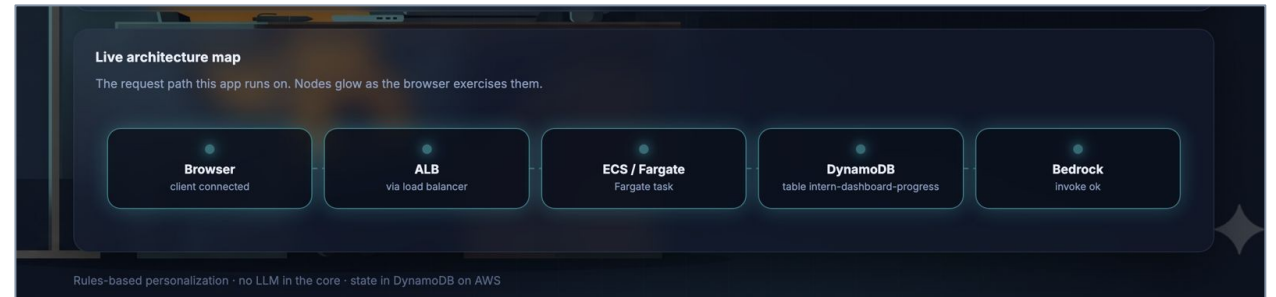
AI Coach — raise the bar

Architecture



How a request flows

- It hits CloudFront first — HTTPS and WAF happen at the edge.
- Static files like CSS and images come from the edge cache.
- Dynamic requests go to the ALB, which only accepts CloudFront.
- The ALB hands it to a healthy Fargate task in one of two AZs.
- The app reads/writes DynamoDB over a private VPC endpoint.
- And reaches Bedrock for coaching through the NAT gateway.



A closer look at ECS

► LIVE · open ECS console → Tasks

- My web service keeps two tasks running at all times.
- Each task is two containers — my app plus a telemetry sidecar.
- The sidecar is non-essential, so if it dies the app keeps serving.
- Reminders and analytics are separate jobs that run and then exit.
- ECS treats 'desired count' as a promise it constantly keeps.

Scheduled jobs & alerts

- EventBridge triggers a reminder task on a schedule.
- It emails each intern their due-soon items via SNS.
- A separate analytics task emails a weekly cohort summary.
- Both run, do one job, and exit — no always-on cost.

Intern Dashboard reminders — 2026-07-24



○ AWS Notifications <no-reply@sns.amazonaws.com>

Friday, July 24, 2026 at 5:01PM

To: ○ Mokom, Larry

These interns have overdue or due-soon items to complete:

- Larry (Larry): 18 item(s) need attention before end of day — Wk8: EKS Intermediate (cluster creation, RBAC, IRSA & Pod Identity), Wk9: EKS Intermediate cont. (storage, logs, troubleshooting), Wk8 Labs: IRSA, node compute (managed node groups + Fargate profiles), Wk9 Labs: EBS CSI persistent storage, container logging with CloudWatch, Pass the ECS & EKS assessments (due Jul 24)
- mokoml (mokoml@amazon.com): 3 item(s) need attention before end of day — Deliver the 10-minute capstone presentation + live demo, Project Tensei: build the working prototype, Earn AWS Solutions Architect Associate (SAA) - raise the bar
- me (me): 5 item(s) need attention before end of day — Deliver the 10-minute capstone presentation + live demo, Project Tensei: write the 1-page vision document, Project Tensei: build the working prototype, Deploy on both platforms (ECS + EKS) or both launch types, Earn AWS Solutions Architect Associate (SAA) - raise the bar

Log in to catch up: <https://d1tdk7tryjck3a.cloudfront.net>

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:868962733067:intern-dashboard-alerts:ea805261-2f9b-4796-bb0c-ec51e97862ef&Endpoint=mokoml@amazon.com>

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Weekly Intern Cohort Analytics



○ AWS Notifications <no-reply@sns.amazonaws.com>

Friday, July 24, 2026 at 12:04PM

To: ○ Mokom, Larry

Weekly Intern Cohort Analytics — 2026-07-24

Interns tracked: 3

Average 'meet the bar' completion: 79%

On track: 2 | At risk: 0 | Behind: 1

Cohort is most behind on:

- Wk8: EKS Intermediate (cluster creation, RBAC, IRSA & Pod Identity) (1 interns overdue)
- Wk9: EKS Intermediate cont. (storage, logs, troubleshooting) (1 interns overdue)
- Wk8 Labs: IRSA, node compute (managed node groups + Fargate profiles) (1 interns overdue)
- Wk9 Labs: EBS CSI persistent storage, container logging with CloudWatch (1 interns overdue)
- Deployed on ECS or EKS, accessible via a URL (1 interns overdue)

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<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:868962733067:intern-dashboard-alerts:ea805261-2f9b-4796-bb0c-ec51e97862ef&Endpoint=mokoml@amazon.com>

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Why these choices

- ECS over EKS — no Kubernetes control plane to manage at my scale.
- CloudFront gives me HTTPS, WAF, and caching — and lets me hide the ALB.
- DynamoDB fits a simple lookup by intern — fast, nothing to patch.
- Fargate means there's no server for me to size or maintain.

Staying available

► LIVE • ECS console → Tasks (show 2 AZs)

- Two tasks, one in each Availability Zone.
- If an AZ fails, the ALB routes to the survivor and ECS rebuilds.
- DynamoDB copies every write across three AZs automatically.
- Being honest: it's single-region today — that's my blast radius.
- Next step would be multi-region with Global Tables and Route 53.

Demo: killing a task

► LIVE • run in terminal (CLI)

- I'll show two tasks running, then kill one on purpose.
- The app keeps working — the ALB routes around the dead one.
- Then ECS notices and rebuilds back to two.
- That's 'desired count is a promise' in action.

Demo: scaling under load

► LIVE • run in terminal (CLI)

- I'll throw traffic at the ALB from my terminal.
- Requests-per-target crosses the scaling threshold.
- The service grows from two tasks toward six.
- Scale-in is slow on purpose so it doesn't flap.

A problem I ran into

- At first my ALB was public — open to the whole internet.
- A security scan caught it and pulled the listener down on me.
- I fixed it forward: CloudFront in front for HTTPS and WAF.
- And I locked the ALB to only accept CloudFront and my own IP.

What I'd do next

- At scale, my analytics table Scan would slow down first.
- I'd switch it to a targeted Query on an index instead.
- Add retries with backoff for when Bedrock gets busy.
- And go multi-region to survive a full region outage.

Thank you — questions?

Customer-obsessed · ECS Fargate · auto-scaling + secrets + self-healing